

Koyena Chowdhury

Ph.D. Senior Research Fellow (SRF), CSE, IIT Kharagpur, India

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Education

- Ph.D., Computer Science and Engineering — 8.07/10** 2023 – Present
CNERG Lab, Indian Institute of Technology (IIT) Kharagpur
• Supervisor: Prof. Saptarshi Ghosh
• Thesis Topic: Application of AI in Analyzing and Optimizing Indian Railways
- M.Tech., Computer Science and Engineering — 9.28/10** 2021 – 2023
National Institute of Technology (NIT) Durgapur
• Supervisor: [Prof. Animesh Dutta](#)
• Thesis Topic: Influence Maximization and Ranking of Influential Spreaders in Social Networks
- B.Tech., Computer Science and Engineering — 9.50/10** 2017 – 2021
Academy of Technology (AOT), MAKAUT, Adisaptagram
- Higher Secondary (WBCHSE) — 88.2%** 2017
Baidyabati Charushila Bose Balika Vidyalaya (BCBBV), Hooghly
- Secondary (WBBSE) — 87%** 2015
St. Anthony's High School, Chandannagore, Hooghly

Publications

- Koyena Chowdhury**, Paramita Koley, Abhijnan Chakraborty, and Saptarshi Ghosh. 2025
“RSTGCN: Railway-centric Spatio-Temporal Graph Convolutional Network for Train Delay Prediction.”
Under Review at IEEE Transactions on Intelligent Transportation Systems (T-ITS) (Available on ArXiv) [[Paper](#)]
- Koyena Chowdhury**, Amit Paul, and Animesh Dutta. 2025
“CancelOut CommGCN (CCGCN): Influence Maximization on Social Networks based on Communities.”
In Proceedings of the 13th ACM IKDD International Conference on Data Science (CODS), pp. 27-34. 2025. [[Paper](#)]
- Koyena Chowdhury**, Amit Paul, Animesh Dutta. 2023
“CancelOut GCN Diffusion (CoutGCN): Finding the Influential Spreaders to Diffuse Information.”
15th International Conference on Communication Systems & Networks (COMSNETS), IEEE, 2023. [[Paper](#)]
- Anju Bhuiya, **Koyena Chowdhury**, Suman Nandi, Animesh Dutta. 2023
“Semi-Global Circular Centrality to find Influential Spreaders.”
15th International Conference on Communication Systems & Networks (COMSNETS), IEEE, 2023. [[Paper](#)]

Ongoing Projects

- Indian Railway Evolution (1991–2026): Temporal network analysis and growth-model simulation.
- Kharagpur Platform Allocator: AI-assisted train–platform assignment under operational constraints.
- Multi-Country Railway Networks: Comparative analysis of railway network structure and delay dynamics.

Research Interests

- Graph Neural Networks
- Transportation Analytics
- Spatio Temporal Foundation Models
- Complex Network Analysis
- Graph Theory and Social Networks
- Influence Maximization and Node Detection

Technical Skills

Programming: Python, C, C++, Java, SQL

Libraries & Frameworks: PyTorch, TensorFlow, Keras, MXNet, Scikit-learn, Pandas, NumPy, NetworkX, Seaborn

AI/ML & Research Areas: GNNs, LLMs, Spatio-Temporal Learning, Time-Series Forecasting, Complex Network Analysis, Influence Maximization, Transportation Analytics, Network Growth Modeling

Teaching Experience

- Hands-on Sessions for Directors-level Govt. Officials from ministries – AI & Big Data, IIT Kharagpur 2026
- Hands-on Training Sessions for ISS Probationers – AI & Big Data, IIT Kharagpur 2026
- Invited Resource Person, ACM Winter School – Responsible AI, XIM University, Bhubaneswar 2025
- Hands-on Workshop Sessions for IOCL Govt. Officials – Advanced AI & ML Workshop, IIT Kharagpur 2025
- Lab Sessions, Mid-Career Training Program for ISS Officers – AI & Big Data, IIT Kharagpur 2024
- Teaching Assistant – NLP '26, NLP '25, NLP '24, DA '25, DA '24, IIT Kharagpur
- Teaching Assistant – Distributed Systems '23, DBMS Lab '23, Data Structures '22, NIT Durgapur

Leadership and Service

- RS Representative, CSE – Certificate of Appreciation by Dean (FoE&A), IIT Kharagpur 2025 – 2026
- Social Media Manager, CNeRG Lab ( LinkedIn, X Twitter) 2024 – Present
- Mentored M. Tech., B. Tech., and Dual Degree Students, IIT Kharagpur

Awards and Honors

- Travel Grant Award – CODS Conference, IISER Pune 2025
- Travel Grant Award – IndoML Conference, IIT Bombay 2023
- Travel Grant Award – COMSNETS Conference, Bangalore 2023
- Best Project Award, M.Tech., CSE, NIT Durgapur 2023
- 2nd Rank in Ph.D. Comprehensive Exam, IIT Kharagpur 2023
- 2nd Rank in M.Tech., CSE, NIT Durgapur 2023
- 2nd Rank in B.Tech., CSE, Academy of Technology 2021
- All India Score – 527, Percentile – 96.42 in GATE 2021
- ICICI Prudential Chairman Circle Child Scholarship 2017
- 2nd Rank in Higher Secondary, Science Dept., BCBBV 2017

Academic Activities

- Poster Presentation and Cultural Program Participation – CSE Research Scholars' Day, IIT Kharagpur 2026
- Poster Presentation – CSE Research Scholars' Day, NIT Durgapur 2023
- B.Tech. Project – Fake Review Detection using Machine Learning and Deep Learning 2021
- Industrial Training at NIVT – Fake News Detection using Machine Learning 2020
- Qualified 2nd Round – TCS CodeVita 2019
- Qualified 2nd Round – KPIT Sparkle, Pune 2019

References

Prof. Saptarshi Ghosh

Ph.D. Supervisor

Associate Professor, CSE, IIT Kharagpur

Email: saptarshi@cse.iitkgp.ac.in

Prof. Abhijnan Chakraborty

Ph.D. Collaborator

Assistant Professor, CSE, IIT Kharagpur

Email: abhijnan@cse.iitkgp.ac.in